

ISBA 2411 | FINAL TEAM PROJECT

MSBA Job Matching

Grounded NLP decision support
for career advisors

Search | Fit | Evidence | Human review

Yitong Wang | Zhike Yan



The advisor must separate fit, uncertainty, and risk

A useful system improves a specific decision: forward, hold, or investigate.

FORWARD

Strong entry-level analytics fit supported by title, skills, and location evidence.

HOLD

Senior, engineering-heavy, or clearly misaligned roles require a conservative screen.

INVESTIGATE

Incomplete fields or missing authorization language require human follow-up.

Role fit and work authorization are related in workflow, but they are not the same prediction task.

Label quality, not data volume, is the bottleneck

The final model uses the full training split and audits all 785,741 source rows.

785,741

original public job
records

100,000

stratified project
sample

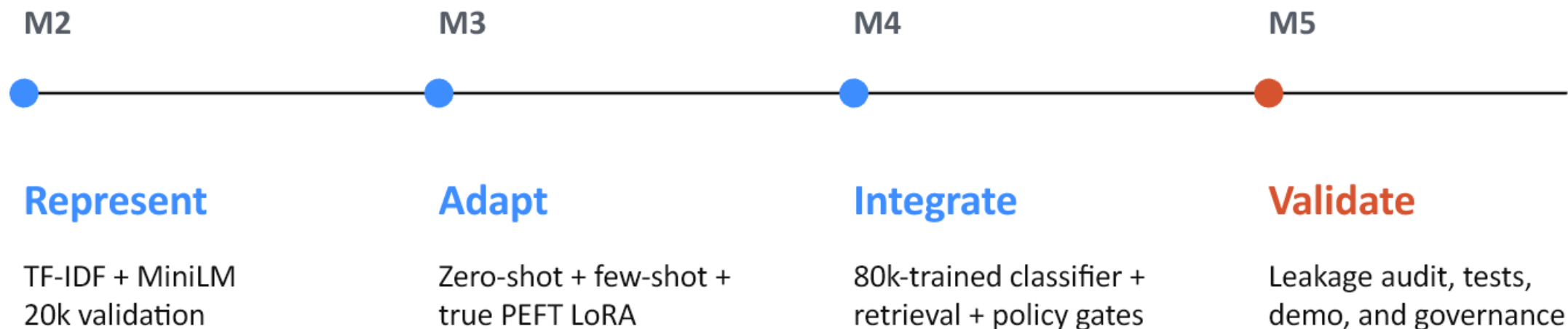
77,135

deduplicated training
rows

230

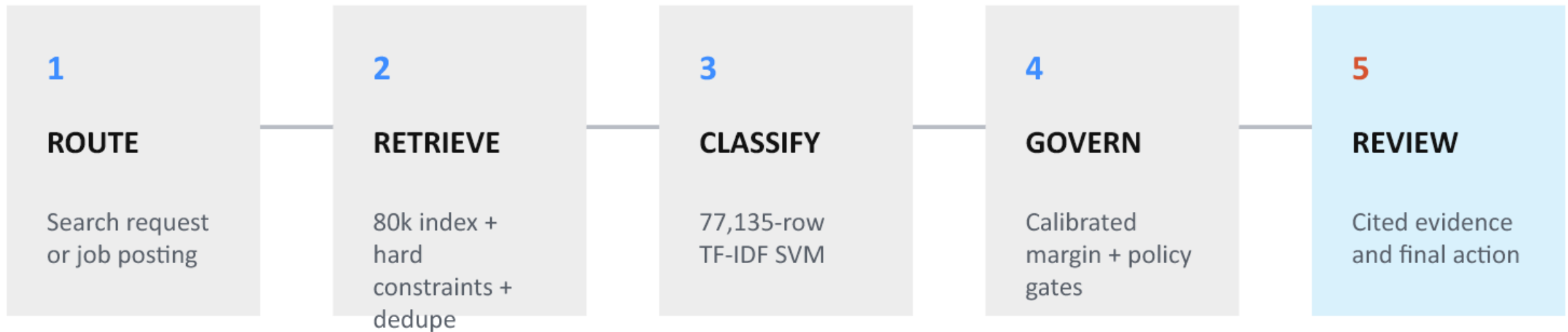
unique authorization
candidates

Each milestone changed the final technical decision



Two input modes prevent a subtle but costly task error

Search requests retrieve candidates; only job postings receive a role-fit label.



A high role-fit score can never override an explicit authorization restriction.

A real search returns constrained, inspectable evidence

```
PS> # output saved to outputs/final_demo_output.json
```

```
input_mode: search
```

```
posting_label: N/A (the request is not a posting)
```

1. Data Analyst | HireMatch | United States | high_fit
2. Data Analyst | HMP Global | United States | high_fit
3. Data Analyst | AMIRI | United States | high_fit

```
review_required: True
```

```
authorization: not available in public source
```

```
Do not infer CPT, OPT, or sponsorship from this public dataset. Authorization evidence must be reviewed separately.
```

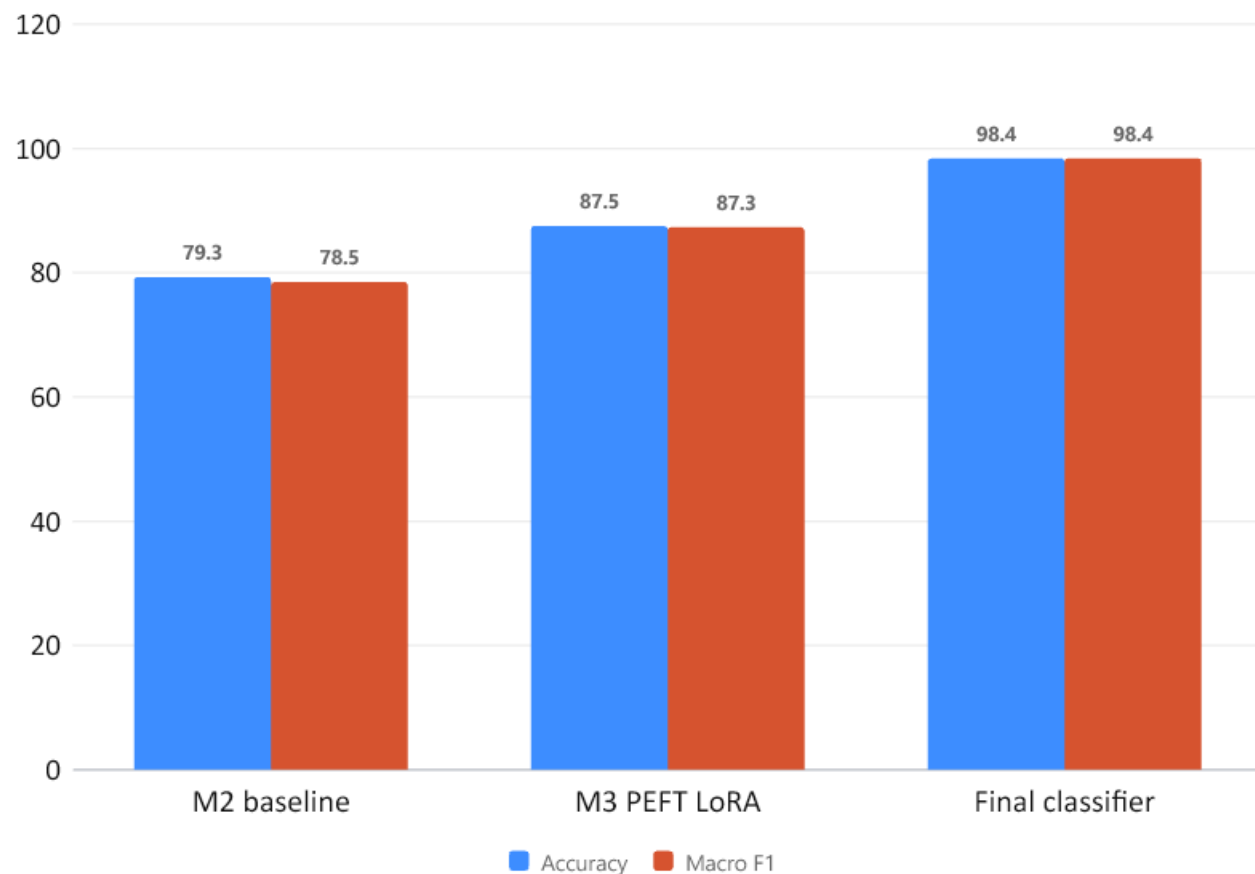
6 audit queries -> 36 results -> 0 hard-constraint violations -> 0 duplicate company-title pairs

High role fit never overrides an explicit restriction

```
PS> # output saved to outputs/final_edge_case_output.json
model_role_fit: high_fit (margin 4.422)
authorization_evidence: explicit source text: no OPT/CPT
review_required: True
recommended_action: Hold - explicit authorization restriction requires advisor verification
top cited posting: Jr. Data Analyst (No OPT/CPT) at Winorbit Technology
RESULT: high role fit is preserved, but the workflow action is HOLD.
```

No OPT/CPT -> HOLD. Missing core fields -> UNCLEAR + REVIEW.

More data worked; extra complexity did not



98.4% fixed-set macro F1

97.9% novel-text macro F1

0 unsupported
authorization claims

The final 3.8 MB model trains in under one minute.

The remaining risk is label validity, not model capacity

FIXED 4K ERRORS BY TRUE LABEL



DEPLOYMENT RULE

Treat 0.984 as weak-label agreement, not advisor accuracy.

Operational validation:
91.6% selective coverage
99.47% retained accuracy
36 search results, 0 violations

Ready for a supervised Career Center pilot

Deploy with guardrails

Prioritize, do not auto-reject
Keep advisor confirmation
Measure time saved and overrides

Batch review with auditable policy reasons

OPERATIONS

```
PS> python -m msba_job_matcher.batch --input demo/batch_review_input.csv
input_row | mode | label | review | action
1 | search | not_applicable_search_query | True | Advisor review before forwarding
2 | posting | high_fit | True | Hold: explicit authorization restriction requires advisor
verification
3 | posting | unclear | True | Investigate: advisor review required
Each row preserves retrieved evidence, policy reasons, and the full JSON result.
One model/index load processes the complete advisor queue.
```

98.4% = weak-label agreement

Next: label the 1,000-row advisor queue and measure real workflow value.